0.1 Continuum Mechanics and Kinematical Relations

<INSERT CLASSIC CONTINUUM MECHANICS IMAGE OF THE POTATO BUT IN ORTHONORMAL BASIS>

According to continuum mechancis, a body $\mathcal{B}$ is defined as a manifold consisting of finite number of particles. This body, $\mathcal{B}$, is also assumed to be a σ - finite measure space - this implies that each subset of $\mathcal{B}$ is associated with a non-negative measure, which in this case would be the mass distribution. This manifold is assumed to be smooth, sufficiently differentiable, and the configuration of the particales at a given time can be mapped one-to-one onto Euclidean space, $\varphi : \mathcal{B} \to \mathbb{E}^3$, thus assigning a position vector to each particle of the body in $\mathbb{E}^3$.[3.1][3.2] For simplicity, an orthogonal cartesian basis is assumed in this section as the formulation will be based on isoparametric interpolations - which rely on an orthogonal basis. A more general description using curvilinear coordinates is covered in Section **??**. In order to ease the description of motion and deformation, it is convenient to assume a *reference configuration* at an arbitrary time, $t = t_0$. In this configuration, the body is assumed to be unloaded, undeformed, and unstressed. For an arbitrary point on the body, P_0, a position vector can be defined with respect to the origin of the basis,

$$\begin{cases} \mathbf{X} &= \boldsymbol{\varphi}(X, t_0), \\ \mathbf{X} &= \mathrm{X}^i\,\mathbf{E}_\mathrm{i}\,, \text{ where } \mathrm{X}^i : \mathrm{X}^1, \mathrm{X}^2, \mathrm{X}^3 \end{cases} \tag{1}$$

and $\mathbf{E}_\mathrm{i}$ are unit vectors along the X^i axes.

At any other time, t, the configuration of the body will be referred to as the *current configuration*. This mapping of the reference configuration into the current configuration is assumed to be so that all points in both configurations are mapped one-to-one. The position vector of the point P_0, that has now moved due to external forces to a position P, can be defined as,

$$\begin{cases} \mathbf{x} &= \boldsymbol{\varphi_t}(\boldsymbol{X}, t), \\ \mathbf{x} &= \mathrm{x}^i\,\mathbf{e}_\mathrm{i}\,, \text{ where } \mathrm{x}^i : \mathrm{x}^1, \mathrm{x}^2, \mathrm{x}^3. \end{cases} \tag{2}$$

Similarly, $\mathbf{e}_\mathrm{i}$ are the unit vectors along the x^i axes.

Two consequences follow the assumption of a cartesian basis system. The dual basis to the unit vectors $\mathbf{E}_\mathrm{i}$ and $\mathbf{e}_\mathrm{i}$ reduces to the original basis, thus making the position of the indices immaterial. And $\mathbf{E}_\mathrm{i} = \mathbf{E}^\mathrm{i} = \mathbf{e}_\mathrm{i} = \mathbf{E}^\mathrm{i}$, thus making the unit vectors and the position of the indices in both the configurations interchangeable without any effects to the maths involved.

The position vector of the point P relative to P_0 is called the displacement vector, $\mathbf{u}$,

$$\mathbf{u} = \mathbf{x} - \mathbf{X} = \mathrm{U}^\mathrm{i}\mathbf{E}_\mathrm{i} = \mathrm{u}^\mathrm{i}\mathbf{e}_\mathrm{i} \tag{3}$$

To study the deformation of the body two vectorial line elements $\mathrm{d}\boldsymbol{X}$, defined as an infinitesimal vector between two points in the reference configuration, and the vectorial line element $\mathrm{d}\boldsymbol{x}$, defined as the infinitesimal vector between the corresponding two points mapped to the current configuration, are used. $\mathrm{d}\boldsymbol{X}$ is mapped to $\mathrm{d}\boldsymbol{x}$ with the help of a second order called the *deformation gradient*.

$$\mathrm{d}\boldsymbol{x} = \mathbf{F}\,\mathrm{d}\boldsymbol{X} \tag{4}$$

$$\mathbf{F} := \mathrm{Grad}\,\mathbf{x} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}^\mathrm{i}} \otimes \mathbf{e}^\mathrm{i} = \frac{\partial \mathrm{x}^\mathrm{i}}{\partial \mathrm{X}^\mathrm{j}}\,\mathbf{e}_\mathrm{i} \otimes \mathbf{e}^\mathrm{j} = \mathrm{F}^\mathrm{i}_{\cdot\mathrm{j}}\,\mathbf{e}_i \otimes \mathbf{e}^j \tag{5}$$

The deformation gradient further allows the transformation of differential quantities from the reference to the current configuration. A differential area element $\mathbf{da}$ in the current configuration can be written in terms of the area element $\mathbf{dA}$ in the reference configuration using Nanson's formula,

$$\mathbf{da} = \mathbf{n}\,da = J\boldsymbol{F}^{-T}\mathbf{N}\,dA = J\boldsymbol{F}^{-T}\mathbf{dA}, \tag{6}$$

where $\mathbf{n}$ and $\mathbf{N}$ are the normal vectors of the area elements $\mathbf{da}$ and $\mathbf{dA}$ respectively, and J is the determinant of the deformation gradient

$$J = \det \boldsymbol{F}. \tag{7}$$

This also enables us to construct a relation between the differential volume elements,

$$dv = J\,dV. \tag{8}$$

The squares of the line elements $\mathbf{dx}$ and $\mathbf{dX}$ are related with the use of the right and left *Cauchy-Green tensors*, $\boldsymbol{C}$,

$$\mathbf{dx} \cdot \mathbf{dx} = \mathbf{dX} \cdot \boldsymbol{C}\mathbf{dX}, \text{ where, } \boldsymbol{C} := \boldsymbol{F}^T\boldsymbol{F}^T, \tag{9}$$

$$\mathbf{dX} \cdot \mathbf{dX} = \mathbf{dx} \cdot \boldsymbol{b}^{-1}\mathbf{dx}, \text{ where, } \boldsymbol{b} := \boldsymbol{F}\boldsymbol{F}^T, \tag{10}$$

which enables us to define the strain measures known as the *Green-Lagrange strain tensor* and the *Almansi strain tensor* respectively

$$\boldsymbol{E} := \frac{1}{2}\left(\boldsymbol{C} - \boldsymbol{I}\right), \tag{11}$$

$$\boldsymbol{e} := \frac{1}{2}\left(\boldsymbol{I} - \boldsymbol{b}^{-1}\right). \tag{12}$$

0.2 Stress and Balance Equations

On a body $\mathcal{B}$, the Cauchy stress vector $\mathbf{t}$ refers to a force per unit area of the deformed surface with a unit normal vector

$$\mathbf{t} = \lim_{\Delta a \to 0} \frac{\boldsymbol{\Delta}\mathbf{p}}{\Delta a}. \tag{13}$$

According to the *Cauchy postulate*, the stress vector $\mathbf{t}$ is only dependent on the unit normal $mathbfn$ and not on any other characteristic of the surface. This mapping between the stress vector and the unit normal can be represented with the help of a second order tensor called the *Cauchy stress tensor*

$$\mathbf{t} = \boldsymbol{\sigma}\mathbf{n}, \text{ where } \boldsymbol{\sigma} = \sigma^{ij}\mathbf{e}_i \otimes \mathbf{e}_j. \tag{14}$$

The fundamental relations of continuum mechanics are the balance relations. They are stated in the following briefly and without proof. The derivations can be found in [3.2] and [3.3].

0.2.1 Balance of Mass

The mass of a system is conserved. This implies that the mass of the body in the reference configuration is equal to the mass of the body in the current configuration.

$$\dot{\boldsymbol{m}} = 0, \tag{15}$$

$$\rho\, dv = \rho_0\, dV\,, \tag{16}$$

where ρ and dv are the mass density and the volume of the body in the current configuration, and ρ_0 and dV are the density and volume of the body in the reference configuration.

0.2.2 Balance of Linear Momentum

The rate of change of linear momentum of a body is equal to the sum of the external forces acting on the body.

$$\begin{cases} \mathrm{div}\boldsymbol{\sigma} + \boldsymbol{f} = \rho\, \ddot{\boldsymbol{x}}, \\ \sigma^{ij}||_j + f^i = \rho\, a^i, \end{cases} \tag{17}$$

where $\boldsymbol{f}$ is the sum of forces, excluding inertia, acting on a body. $\ddot{\boldsymbol{x}}$ and $\boldsymbol{a} = a^i\, \boldsymbol{e}_i$ is the acceleration of a particle with the position vector $\boldsymbol{x}$. The First Piola-Kirchhoff stress tensor can be used to represent the linear momentum balance in the reference configuration.

$$\begin{cases} \mathrm{Div}\mathbf{P} + \boldsymbol{f}_0 = \rho_0\, \ddot{\boldsymbol{x}}, \\ P^{ij}|_j + f_0^i = \rho_0\, a_0^i. \end{cases} \tag{18}$$

In (17) and (18), the differential operators $(\bullet)|_j$ and $(\bullet)||_j$ are the covariant derivatives in the current and the reference configurations respectively.

0.2.3 Balance of Angular Momentum

The rate of change of angular momentum of a body is equal to the resultant moment of the external forces acting on the body.

$$\int_{\mathcal{B}} \rho(\boldsymbol{x} \times \boldsymbol{v}) = \int_{\mathcal{B}} \rho(\boldsymbol{x} \times \boldsymbol{b}) + \int_{\partial\mathcal{B}} (\boldsymbol{x} \times \boldsymbol{t}). \tag{19}$$

This results in the local balance of angular momentum which simply requires the Cauchy stress tensor to be symmetric

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\boldsymbol{T}}, \tag{20}$$

$$\sigma_{ij} = \sigma_{ji}. \tag{21}$$

0.2.4 Balance of Kinetic Energy

An extension of (17), is the balance of kinetic energy which states that the power of external forces is equal to the sum of the stress power and the change in kinetic energy of a body.

$$P = W + \dot{K}, \tag{22}$$

$$\text{where the power of external forces, } P = \int_{\mathcal{B}} \rho\, \overline{\mathbf{b}} \cdot \mathbf{v}\, dv + \int_{\partial\mathcal{B}} \mathbf{t} \cdot \boldsymbol{v}\, da, \tag{23}$$

$$\text{the stress power, } W = \int_{\mathcal{B}} \boldsymbol{\sigma} : \mathbf{d}\, dv, \tag{24}$$

$$\text{and the change in kinetic energy, } \dot{K} = \int_{\mathcal{B}} \frac{1}{2}\rho \boldsymbol{v} \cdot \boldsymbol{v}\, dv. \tag{25}$$

Represented by the Cauchy stress tensor, $\boldsymbol{\sigma}$, and the rate of deformation tensor, $\mathbf{d}$, the stress power can be decomposed further into heat production, $\mathcal{D}$ ad the stored energy density, Ψ, as follows

$$W = \mathcal{D} + \int_{\mathcal{B}} \dot{\Psi}\, dV_0. \tag{26}$$

0.2.5 Conservation of Energy

The First Law of Thermodynamics states that the rate of change of the total energy of a system is equal to the sum of the mechanical power of the external loads and the heat supplied to the system.

$$\dot{E} = P + Q, \tag{27}$$

$$\text{where the total energy, } E = \int_{\mathcal{B}} \frac{1}{2}\rho \boldsymbol{v} \cdot \boldsymbol{v}\, dv + \int_{\mathcal{B}} \rho u\, dv, \tag{28}$$

$$\text{the power of external loads, } P = \int_{\mathcal{B}} \rho\, \overline{\mathbf{b}} \cdot \mathbf{v}\, dv + \int_{\partial\mathcal{B}} \mathbf{t} \cdot \boldsymbol{v}\, da, \tag{29}$$

$$\text{and the heat supplied, } Q = -\int_{\partial\mathcal{B}} \mathbf{q} \cdot \mathbf{n}\, da + \int_{\mathcal{B}} \rho\, r\, dv \tag{30}$$

where $\mathbf{q}$ and $\mathbf{n}$ are the heat flux vector and the surface normal respectively, and r is the distributer inner heat source.

0.3 Constitutive Relations, Material Model

Constitutive relations are the properties of the material under investigation. They characterise the material response of a body [3.2] and are necessary to complete the system of differential equations for the solution of boundary value problems [3.3]. Materials in the real world can exhibit incredibly diverse behaviours making it necessary for the constitutive relations to have a degree of approximation involved in their formulation. However, the relations must agree with the observed experimental behaviours. Along with the Second Law of Thermodynamics, there are three general principles that all constitutive relations must satisfy, known as *Noll's axioms*. These are the *principle of determinism*, which states that the stress in a particular point can be determined by the stress history at that point; the *principle of local action*, that states that the stress at a particular point is affected by the motion of the particles in immediate neighbourhood of that point; and the *principle of material objectivity*, which states that the choice of coordinate system should not affect the material properties. Hyperelastic materials can be characterised by having zero heat dissipation. Thus the stress power (26) reduces to

$$W = \int_{\mathcal{B}} \dot{\Psi}\, dV_0. \tag{31}$$

Comparing equations (22) and (31), according to [3.3],

$$\begin{aligned} \dot{\Psi} &= \mathrm{J}\boldsymbol{\sigma} : \boldsymbol{d} = \boldsymbol{S} : \frac{1}{2}\dot{\boldsymbol{C}} = 2\frac{\partial \Psi}{\partial \boldsymbol{C}} : \frac{1}{2}\dot{\boldsymbol{C}}, \\ \therefore\ \boldsymbol{S} &= 2\frac{\partial \Psi}{\partial \boldsymbol{C}}. \end{aligned} \tag{32}$$

Neo-Hookean materials are examples of hyperelastic materials. The model can be used to approximate the behaviour of rubber like materials fairly accurately for strains upto 20% [3.4]. Given a Neo-Hookean material model defined as,

$$\boldsymbol{\sigma} = \frac{\mu}{\mathrm{J}}\left(\mathbf{b} - \mathbf{I}\right) + \frac{\Lambda}{\mathrm{J}} \ln \mathrm{J}\mathbf{I} \tag{33}$$

The expression for the compliance matrix in the reference configuration can be written as

$$\mathbb{C} = 4\frac{\partial^2 \Psi}{\partial \boldsymbol{C} \partial \boldsymbol{C}} = 2\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{C}} \tag{34}$$

where $\mathbf{S}$ is the second Piola-Kirchhoff stress tensor which is defined as,

$$\boldsymbol{S} = \mathrm{J}\boldsymbol{F}^{-\mathbf{1}}\boldsymbol{\sigma}\boldsymbol{F}^{-\boldsymbol{T}} \tag{35}$$

The given material model can be written using the Second Piola Kirchhoff Stress Tensor.

$$\boldsymbol{S} = \mu\left(\boldsymbol{I} - \boldsymbol{C}^{-\mathbf{1}}\right) + \Lambda \ln \mathrm{J}\boldsymbol{C}^{-\mathbf{1}} \tag{36}$$

Based on (36) and (34), the explicit expression for the compliance matrix $\mathbb{C}$ is derived as follows:

$$\mathbb{C}^{\mathrm{i.a.}}_{.\mathrm{j.b}} = 2\left(-\mu + \Lambda \ln \mathrm{J}\right)\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{a}}\left(\mathrm{C}^{-1}\right)^{\mathrm{b}}_{.\mathrm{j}} + \Lambda\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\left(\mathrm{C}^{-1}\right)^{\mathrm{a}}_{.\mathrm{b}} \tag{37}$$

Applying Push-Forward to (37) to transform $\mathbb{C}$ to the current configuration,

$$\mathbb{c} = 2\left(-\mu + \Lambda \ln J\right)\delta^p_{.s}\delta^t_{.w}\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w + \frac{\Lambda}{2}\delta^p_{.w}\delta^s_{.t}\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_s \otimes \boldsymbol{e}^t \tag{38}$$

0.4 Weak Form

Using the strong form of the linear momentum balance, (17), the weak form can be derived using a test function as described in Section **??** and the contributions can be separated into *internal virtual work* and *external virtual work*. A derivation of the weak form using general curvilinear coordinates can be found in Section **??**, which can be reduced to obtain the following

$$\underbrace{\int_{\mathcal{B}} \mathrm{grad}(\boldsymbol{w}) : \boldsymbol{\sigma}\, dv}_{\text{internal virtual work}} - \underbrace{\int_{\mathcal{B}} \boldsymbol{w} \cdot \rho\bar{\boldsymbol{b}}\, dv - \int_{\partial_t \mathcal{B}} \boldsymbol{w} \cdot \bar{\boldsymbol{t}}\, da}_{\text{external virtual work}} = 0 \tag{39}$$

where the contributions of the internal virtual work and external virtual work can be defined as G_{int} and G_{ext} respectively.

0.5 Discretisation

The reference configuration $\mathcal{S}_0$ is discretised into a set of finite elements, Ω^e_0 that are defined by nodal points $\boldsymbol{X}$. Thus, the current configuration is composed by finite elements Ω^e with nodal points $\boldsymbol{x}$ that are related to $\boldsymbol{X}_I$ through relations described in 0.1.

After the discretisation of the domain into finite elements, the shape functions can be used to discretise the nodal variables in the reference and current configurations $\boldsymbol{X}$ and $\boldsymbol{x}$, the primary field variables $\boldsymbol{u}$, and the test functions $\boldsymbol{w}$ within each finite element can be discretised using the shape functions according to the isoparametric concept as described in Section **??**. Given a finite element domain, Ω, and shape functions, $\boldsymbol{N}^e$,

$$\boldsymbol{X} \approx \boldsymbol{X}^h = \sum_{k=1}^{n_{en}} N^e_k X^e_n = \mathbf{N}^e\mathbf{X}^e, \tag{40}$$

$$\boldsymbol{x} \approx \boldsymbol{x}^h = \sum_{k=1}^{n_{en}} N^e_k x^e_n = \mathbf{N}^e\mathbf{x}^e, \tag{41}$$

$$\boldsymbol{u} \approx \boldsymbol{u}^h = \sum_{k=1}^{n_{en}} N^e_k u^e_n = \mathbf{N}^e\mathbf{u}^e, \tag{42}$$

$$\boldsymbol{w} \approx \boldsymbol{w}^h = \sum_{k=1}^{n_{en}} N^e_k w^e_n = \mathbf{N}^e\mathbf{w}^e, \tag{43}$$

where the summation is carried over the $n_n e$ nodes of the element. Using the above given definitions, and the weak form of the momentum balance (39), the discretised form of the weak form in its internal virtual work contributions can be given as,

$$G_{int} \approx G^h_{int} = \sum_{k=1}^{n^e_n} G^k_{int}, \text{ where } G^e_{int} = \int_{\Omega^e} \mathrm{grad}(\boldsymbol{w}) : \boldsymbol{\sigma}\, dv$$

$$\text{and, } G^e_{int} = \boldsymbol{w}^T_e \int_{\Omega^e} \mathbf{B}^{\boldsymbol{T}}_{\boldsymbol{e}}\boldsymbol{\sigma}_{\boldsymbol{v}}\, dv =: \boldsymbol{w}^T_e \mathbf{f}^e_{int} \tag{44}$$

As the Cauchy stress tensor is symmetric in this framework, to ease computation we can take use of the Voigt notation. Thus, $\boldsymbol{\sigma_v}$ is the stress tensor written in the Voigt notation,

$$\boldsymbol{\sigma_v} = \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{31} \\ \sigma_{12} \end{Bmatrix} \tag{45}$$

And the matrix $\boldsymbol{B}_e$ is defined as,

$$\boldsymbol{B}_e := [\boldsymbol{B_1}, \boldsymbol{B_2}, \ldots, \boldsymbol{B_n}],$$

$$\boldsymbol{B}_A = \begin{bmatrix} N_{A,x_1} & 0 & 0 \\ 0 & N_{A,x_2} & 0 \\ 0 & 0 & N_{A,x_3} \\ 0 & N_{A,x_3} & N_{A,x_2} \\ N_{A,x_3} & 0 & N_{A,x_1} \\ N_{A,x_2} & N_{A,x_1} & 0 \end{bmatrix} \tag{46}$$

Similarly, the external virtual work can be discretised,

$$G_{\text{ext}} \approx G^h_{\text{ext}} = \sum_{k=1}^{n_n^e} G^e_{\text{ext}},$$
$$\text{where, } \; G^e_{\text{ext}} = \boldsymbol{w}_e^T \left[\int_{\Omega^e} \boldsymbol{N}^T \rho \bar{\boldsymbol{b}} \, dv + \int_{\Gamma^e} \boldsymbol{N}^T \bar{\boldsymbol{t}} \, da \right] =: \boldsymbol{w}_e^T \mathbf{f}^e_{\text{ext}} \tag{47}$$

Assembly of these force vectors, $\mathbf{f}^{\text{e}}_{\text{int}}$ and $\mathbf{f}^{\text{e}}_{\text{ext}}$, based on the element connectivity results in the discretised weak form,

$$\boldsymbol{w}^T [\mathbf{f}_{\text{int}} - \mathbf{f}_{\text{ext}}] = \mathbf{0}. \tag{48}$$

0.6 Newton Raphson Method and Linearisation

The Newton-Raphson method is an iterative method that can be used to approximate the roots of a nonlinear system of equations. The algorithm subdivides a nonlinear system into linearised sub-problems and arrives at the global solution through the sequential solution of the sub-problems. A given system of nonlinear equations,

$$\boldsymbol{f}(\boldsymbol{u}) = \mathbf{0}, \tag{49}$$

can be linearised using the Taylor series expansion as,

$$\boldsymbol{f}(\boldsymbol{u} + \boldsymbol{\Delta u}) = \boldsymbol{f}(\boldsymbol{u}) + D\boldsymbol{f}(\boldsymbol{u}) \cdot \boldsymbol{\Delta u}, \tag{50}$$

after having ignored the higher order terms. In (50), $D\boldsymbol{f}(\boldsymbol{u})$ is the directional derivative of $\boldsymbol{f}$ at $\boldsymbol{u}$. $D\boldsymbol{f}(\boldsymbol{u})$ can be written using the general *Gâteaux* derivative. However, if $D\boldsymbol{f}(\boldsymbol{u})$ is linear in $\boldsymbol{\Delta u}$, it can be written using the *Frechét* differential of $\boldsymbol{f}$ at $\boldsymbol{u}$ as,

$$D\boldsymbol{f}(\boldsymbol{u}) := \boldsymbol{f\prime}(\boldsymbol{u}) \cdot \boldsymbol{\Delta u}, \text{ where, } f'(u)_{ij} = \frac{\partial f_i}{\partial u^j}(u). \tag{51}$$

In (51), $\boldsymbol{f\prime}(\boldsymbol{u})$ can be called the Jacobian or the tangent matrix.

Since the test functions in (48) are arbitrary, the system of equations reduce to

$$\mathbf{f} = \mathbf{f}_{\text{int}} - \mathbf{f}_{\text{ext}} = \mathbf{0}. \tag{52}$$

To be able to use the Newton method to solve (52), the tangent matrix related to $\boldsymbol{G}_{\text{int}}$ needs to be calculated. Linearising $\boldsymbol{G}_{\text{int}}$ at the element level,

$$D\boldsymbol{G}^e_{int}(\mathbf{u}, \boldsymbol{\Delta u}) = \mathbf{w}_e^T \int_{\Omega_0^e} \mathbf{N}^T_{,i} \sigma_{ij} \mathbf{N}_{,j} \, dv \; \boldsymbol{\Delta u}_e + \mathbf{w}_e^T \int_{\Omega_0^e} \mathbf{B}^T \mathbb{c}_v \mathbf{B} \, dv \; \boldsymbol{\Delta u}_e \tag{53}$$

where

$$\mathbf{N}_{,i} = \frac{\partial \mathbf{N}}{\partial x^i} \tag{54}$$

and $\mathbb{c}_v$ is the compliance matrix as derived in equation (38) written in the Voigt notation.